are $\binom{n+(k-1)}{k-1}$ ways to do this.

Homework: Prove it.

17th November Lecture-23

Recap

1) Combinatorics
   a) Bijection rule
   b) Product rule
   c) Sum rule
   d) Generalized product rule
   e) Division rule
   f) Subset rule

6.7 Counting sequences with repetition

An alternate way to count the number of k-element subsets of

a set A of n elements is as follows: Consider the pair, where the first component in this pair is a k-element subset of A & the second element of this pair is the set of remaining $(n-k)$ elements of A. So, the subset rule can be viewed as counting the number of possible ways to split A into such pairs. This can be generalized as follows:

$A \downarrow (B, C)$, $B \subseteq A$, $|B| = k$ & $C = A - B$.

Let $n \in \mathbb{N} - \{0\}$ & $k_1, \cdots, k_m \in \mathbb{N}$ such that $k_1 + \cdots + k_m = n$. Let B be a set containing n elements. A $(k_1, \cdots, k_m)$-split

of B is the following sequence

$(B_1, \ldots, B_m)$, → this forms a partition of B

where $B_1, \ldots, B_m$ are disjoint subsets of B and for every $i \in \{1, \ldots, m\}$, $|B_i| = k_i$. To count the number of $(k_1, \ldots, k_m)$-splits of B, we use the same idea as used in the previous section. In particular, we map every bijective function $g : \{1, \ldots, n\} \to B$ into a $(k_1, \ldots, k_m)$-split of B, where the first set B_1 contains the first k_1 images of g, i.e., $B_1 = \{g(1), \ldots, g(k_1)\}$, B_2

contains the next k_2 elements, i.e., $B_2 = \{g(k_1+1), \cdots, g(k_2)\}$ & so on. Clearly, this map is as $(k_1! k_2! \cdots k_m!)$-to-1 function & hence by division rule, we get that the number of $(k_1, \cdots, k_m)$-splits of B is equal to $\frac{n!}{k_1! k_2! \cdots k_m!}$. This quantity is called the <u>multinomial coefficient</u> & is denoted as $\binom{n}{k_1, \cdots, k_m}$.

Definition [Multinomial coefficient]:

Let $n, m, k_1, \cdots, k_m \in \mathbb{N}$, $m \neq 0$, $n \neq 0$ be such that $k_1 + \cdots + k_m = n$. Then the multinomial coefficient with respect to $k_1, \cdots, k_m$, denoted $\binom{n}{k_1, \cdots, k_m}$, is defined as

$$\binom{n}{k_1, \cdots, k_m} := \frac{n!}{k_1!\, k_2! \cdots k_m!}$$

From the above discussion, we get the following.

- **The subset-split rule**: The number of $(k_1, \cdots, k_m)$-splits of an n-element set is equal to

$$\binom{n}{k_1, \ldots, k_m}.$$

Question 1: Roll a die 8 times. In how many sequences do we see exactly two 1's, three 2's, and three 3's?

Answer: Here $n=8$, $k_1=2$, $k_2=3$, $k_3=3$. Then, there are $\binom{8}{2,3,3}$ sequences containing exactly two 1's, three 2's & three 3's.

Question 2: Let $n \in \mathbb{N}$ such that $n \geq 5$. Count the number of n-bit 0,1 sequences which

start either with 1 or start with 0 and contain exactly 4 1's.

Answer: Let A be the set of all n-bit 0,1 sequences that start with 1 and B be the set of all n-bit 0,1 sequences that start with 0 & contain exactly four 1's. We want to find $|A \cup B|$. Note that A & B are disjoint. Hence, by the sum rule

$$|A \cup B| = |A| + |B|.$$

As every sequence in A start

with 1, note that

$$A = \{1\} \times \{0,1\}^{n-1}.$$

Hence, by the product rule, $|A| = 2^{n-1}$. On the other hand, every sequence in B starts with 0 & contains exactly four 1's. This is basically the set of sequences, where the first bit is fixed to 0 & the remaining $(n-1)$ bits contain exactly four 1's. Then, by the subset rule, $|B| = \binom{n-1}{4}$.

Hence, $|A \cup B| = 2^{n-1} + \binom{n-1}{4}$.

6.8 The Bookkeeper Rule

We can generalize the method of counting n-bit 0,1 sequence with k 1's to counting n-element sequences over a set containing more than two elements.

Let us look at an example.

Question: How many sequences can be formed by permuting the letters of the word BOOKKEEPER?

An example of such

sequences are OOBKEEKPER.

Note that BOOKKEEPER contains 1 B, 2 O, 2 K, 3 E, 1 P & 1 R. We want to understand the number of sequences formed by permuting the words of BOOKKEEPER. This immediately leads to a bijective function between the set of permutations of BOOKKEEPER and the set of all (1, 2, 2, 3, 1, 1)-splits of

$\{1, 2, \cdots, 10\}$ (the total number of letters in BOOKKEEPER is 10). This function maps a permutation of BOOKKEEPER to the sequence of the set of positions where each of the different letters in B, O, K, E, P, R occur in the chosen permutation.

for example, suppose we look at the permutation PEBEOKOEKR of BOOKKEEPER. This permutation maps to the set

1 2 3 4 5 6 7 8 9 10

$\{\{3\}, \{5,7\}, \{6,9\}, \{2,4,8\}, \{1\}, \{10\}\}$

where the first subset corresponds to B, second to O, third to K, fourth to E, fifth to P & sixth to R.

From the bijective function between the set of permutations of BOOKKEEPER & the set of all (1,2,2,3,1,1)-splits of $\{1,2,\ldots,10\}$ mentioned above, & the subset split rule, we get that there are $\binom{10}{1,2,2,3,1,1}$

possible permutations of BOOKKEEPER.

This is generalized as follows.

The Bookkeeper Rule: Let $a_1, \cdots, a_m$ be distinct elements. The number of sequences with k_i appearances of a_i for every $i \in \{1, \cdots, m\}$ is equal to

$$\binom{k_1 + k_2 + \cdots + k_m}{k_1, k_2, \cdots, k_m}.$$

Example: Suppose we plan to visit Puri 4 times, Varanasi three times, Udupi five times

& Dwarka two times in a year.
How many possible ways
all there to visit these cities
in a year?

Answer $\binom{14}{4,3,5,2}$.

6.9 Balls & Bins

verify

Question 1: Let $n, m \in \mathbb{N}$, $m \leq n$.

How many ways are there
to put n distinct balls in
m distinct bins?

Answer: m^n ways. (HW)

Que 2: Let $m, n \in \mathbb{N}$, $m \leq n$. verify

How many ways are there to put n identical balls in m distinct bins?

Answer: Suppose we have n unlabelled balls & m distinct bins. We interpret these balls as 0s & denote n balls with n 0s. Now, we put m-1 1's arbitrarily between these n zeros. We interpret it as follows: every 0 to the left of the first

1 in this sequence of $(n+m-1)$ bits corresponds to the 1st bin, then every 0 between the first & the second 1 in this sequence corresponds to the 2nd bin & so on.

We know that there are $\binom{n+m-1}{m-1}$ such sequences.

0 0 | 0 0 | 0

Homework: Let $m, n \in \mathbb{N}$, $m \leq n$. verify

How many ways are there to put n identical balls in m distinct bins such that

each bin contains at least 1 ball?

6.10 The Binomial theorem.

A binomial is a sum of two terms, like $a+b$. Let a, b be two elements, $\wedge i, j \in \mathbb{N}-\{0\}$. Then $a^i b^j$ represents the sequences (or products) where the leftmost i elements are a & the remaining j elements are b.

Theorem [Binomial theorem]:
Let $n \in \mathbb{N}-\{0\}$. For every $a, b \in \mathbb{R}$, we have

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1} b + \cdots$$
$$\cdots + \binom{n}{i} a^{n-i} b^i + \cdots + b^n.$$

Proof: Recall that $(a+b)^n$ is multiplying $(a+b)$ n times. Then, $(a+b)^n$ is the sum of all n-length "words" containing a's & b's. for example,

$$(a+b)^2 = (a+b)(a+b)$$
$$= aa + ab + ba + bb$$
$$= aa + 2ab + bb.$$

Let us pick $i \in \{1, \cdots, n-1\}$ arbitrarily. It follows from the commutativity of multiplication

over $\mathbb{R}$ that all the products of a, b which contain i many as & $(n-i)$ many bs are equal.

we want to count such products & we know that the number of such products is equal to $\binom{n}{i}$. Then, by the sum rule,

$$(a+b)^n = a^n + \binom{n}{1} a^{n-1}b + \cdots + b^n.$$

∎

Observation: Let $n \in \mathbb{N} - \{0\}$ & $i \in \{1, \ldots, n\}$. Then,

$$\binom{n}{i} = \binom{n}{n-i}.$$

6.11 The Inclusion-Exclusion Principle.

Suppose A, B are finite sets such that $A \cap B \neq \emptyset$. How do we find $|A \cup B|$?

Theorem (Inclusion-Exclusion):
Let A, B be arbitrary finite sets. Then,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

19th November Lecture 24

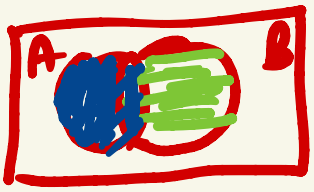
Recap

a) Combinatorics
   1) Bijection rule
   2) Product rule → ×
   3) Sum rule → disjoint ∪
   4) Generalized product rule
   5) Division rule
   6) Subset rule
   7) Subset split rule
   8) Bookkeeper rule
   9) The Binomial theorem

6.11 The Inclusion-Exclusion Principle

Theorem 6.11.1 [Inclusion-Exclusion Principle]: Let A & B be arbitrary finite sets. Then,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$



Proof: Observe that

$$A \cup B = (A - B) \cup B = A \cup (B - A)$$

Note that $(A - B)$ & B (similarly A & $(B - A)$) are disjoint sets. Then, by the sum rule,

$$|A \cup B| = |A - B| + |B| \quad -(i)$$

So, we are left with calculating

$|A-B|$. It is easy to verify (H.W.) that

$$A = (A-B) \cup (A \cap B).$$

Note that $(A-B)$ & $(A \cap B)$ are disjoint sets. Then, by the sum rule,

$$|A| = |A-B| + |A \cap B|$$

$$\Rightarrow |A-B| = |A| - |A \cap B|$$

On plugging this in Equation (1), we get

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Theorem 6.11.2 (Generalized inclusion-exclusion principle):

Let $A_1, \dots, A_n$ be arbitrary finite sets. Then,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j|$$

$$+ \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| + \cdots$$

$$+ (-1)^{n-1} \left|\bigcap_{i=1}^{n} A_i\right|.$$

In other words, we can interpret this as follows:

$\left| \bigcup_{i=1}^{n} A_i \right|$ = the sum of the sizes of individual sets

− the sum of the sizes of two-way intersections

+ the sum of the sizes of three-way intersections

⋮

$+ (-1)^{n-1}$ · the size of the intersection of all sets.

Homework: Prove it by induction on n.

Examples:

1) 20 students in a class like discrete maths, 15 students like algorithms, and 5 students like both. How many students like at least one subject?

Answer: Let A, B be the sets of students liking discrete maths & algorithms. We want to calculate $|A \cup B|$. By the inclusion-exclusion principle,

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$= 20 + 15 - 5 = 30.$$

2) In how many permutations of the set $\{1, \cdots, 9\}$, either 4 and 2, or 5 and 3, or 6 and 1 appear consecutively?

Answer: Let A, B, C be the sets of permutations on $\{1, \cdots, 9\}$ that contain 4 & 2 consecutively, 5 & 3 consecutively, and 6 and 1 consecutively, respectively. For example,

$$\underline{42}13561789 \in A$$
$$124678\underline{53}9 \in B$$
$$2345\underline{61}789 \in C$$

We want to find $|A \cup B \cup C|$. It is not difficult to see that A, B, C are not disjoint sets. Then, by the inclusion-exclusion principle,

$$|A \cup B \cup C| = \overbrace{|A| + |B| + |C|}^{8! \quad 8! \quad 8!} - \overbrace{|A \cap B|}^{7!} - \overbrace{|B \cap C|}^{7!} - \overbrace{|C \cap A|}^{7!} + |A \cap B \cap C|.$$

In every permutation in A, 2 always comes immediately after 4. Thus, we can glue 4 & 2 together and now consider the number of bijective maps

from $\{1,3,5,6,7,8,9,42\}$ to $\{1,\dots,8\}$.
We know that there are $8!$ such maps. Hence, $|A| = 8!$
Similarly, $|B| = |C| = 8!$ Further, note that every element of $A \cap B$ is a bijective function from the set $\{1,6,7,8,9,42,53\}$ to $\{1,\dots,7\}$. Hence, $|A \cap B| = 7!$
Similarly, $|B \cap C| = |A \cap C| = 7!$
Further, $A \cap B \cap C$ contains all bijective functions from $\{7,8,9,42,53,61\}$ to $\{1,\dots,6\}$.

Hence, $|A \cap B \cap C| = 6!$ Thus, we get

$$|A \cup B \cup C| = 3 \cdot 8! - 3 \cdot 7! + 6!$$

6.12 Combinatorial proofs.

A combinatorial proof shows that two different expressions are equal by counting the same set in two different ways.

Let us see an example. Suppose you have a list of n places but you only want to visit k of them. One way is to select k places that you want to

visit and the other way is to select $(n-k)$ places that you don't want to visit. Thus, the number of ways of selecting k stations out of n must be equal to the number of ways of selecting $(n-k)$ places out of n. Hence, we get

$$\binom{n}{k} = \binom{n}{n-k}.$$

This is easy to prove algebraically as both sides are equal to $\frac{n!}{k!\,(n-k)!}$ but in this section we will not use algebraic proofs.

Let us first see the structure of a combinatorial proof & then we will see some examples.

Outline of a combinatorial proof:

A combinatorial proof is an argument that establishes an algebraic fact by relying only on counting principles. The outline of a combinatorial proof is as follows.

Suppose we want to prove that two combinatorial quantities m & n are the same.

1) Define a finite set S.

2) Show that $|S| = n$ by counting one way.

3) Show that $|S| = m$ by counting another way.]

4) Conclude $m = n$.

Claim 6.12.1 [Pascal's triangle inequality]

Let $n \in \mathbb{N} - \{0\}$ and $k \in \mathbb{N}$, $k < n$.

Then,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

We can prove this algebraically, but we only give a combinatorial proof.

Remark: Whenever you are asked to give a combinatorial proof in exam, you can not write an algebraic proof.

Proof: Let $A=\{a_1,\cdots,a_n\}$ be a set of n elements. Let

$$S := \{B \subseteq A : |B| = k\}.$$

We know from the subset rule that $|S| = \binom{n}{k}$. Now, let us see another way of finding $|S|$.

Let $S_1 := \{B \in S : a_1 \in B\}$ and

$S_2 := \{C \in S : a_1 \notin C\}$.

As $S_1 \cap S_2 = \phi$, the sum rule implies

$$|S_1 \cup S_2| = |S_1| + |S_2|.$$

Note that $S = S_1 \cup S_2$. Then,

$$|S| = |S_1| + |S_2|.$$

Note that every set in S_1 contains k elements including a_1.

So, every set in S_1 contains $k-1$ elements from $A - \{a_1\}$.

Then, by the subset rule,

$$|S_1| = \binom{n-1}{k-1}.$$

We know that for every $C \in S_2$, $|C| = k$ & $a_1 \notin C$. Then, by the

subset rule,

$$|S_2| = \binom{n-1}{k}.$$

As $|S| = |S_1| + |S_2|$, we get

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

∎

Claim 6.12.2: Let $m, n \in \mathbb{N} - \{0\}$ such that $n \leq m$. Then,

$$\binom{m+n}{n} = \sum_{i=0}^{n} \binom{n}{i}\binom{m}{n-i}.$$

Proof: Let A be a set containing n distinct English words & m distinct numbers.

Let $S := \{B \subseteq A : |B| = n\}$.

Then, by subset rule,

$$|S| = \binom{m+n}{n}. \quad \text{— (1)}$$

Now, let us see another way of calculating $|S|$. Note that every element of S contains n elements, some of which are English words & remaining are numbers. For $i \in \{0, \cdots, n\}$, let $S_{i,n-i}$ be defined as follows:

$S_{i,n-i} := \{B \subseteq A : B$ contains i English words & $(n-i)$ numbers$\}$.

Let R_i & R_{n-i} be the sets of all i choices of English words & $(n-i)$ choices of numbers. Then, by the subset rule,

$$|R_i| = \binom{n}{i} \quad \text{and} \quad |R_{n-i}| = \binom{m}{n-i}$$

It is not difficult to see that there exists a bijective function from $S_{i,n-i}$ to $R_i \times R_{n-i}$. Then, by the bijection rule,

$$|S_{i,n-i}| = |R_i| \cdot |R_{n-i}|$$
$$= \binom{n}{i} \cdot \binom{m}{n-i} \quad -(ii)$$

Observe that $S_{0,n}, S_{1,n-1}, \cdots, S_{i,n-i}, \cdots, S_{n,0}$ are disjoint sets. Further,

$$S = S_{0,n} \cup S_{1,n-1} \cup \cdots \cup S_{n,0}$$

Then, by the sum rule,

$$|S| = \sum_{i=0}^{n} |S_{i,n-i}|.$$

Then, Equation (ii) implies,

$$|S| = \sum_{i=0}^{n} \binom{n}{i} \binom{m}{n-i}$$

As $|S| = \binom{m+n}{n}$ (Equation (i))

we get

$$\binom{m+n}{n} = \sum_{i=0}^{n} \binom{n}{i} \binom{m}{n-i}.$$

∎

Remark: It is quite challenging to prove the above claim algebraically.

Claim 6.12.3: Let $n \in \mathbb{N} - \{0\}$. Then,

$$\sum_{i=0}^{n} \binom{n}{i} = 2^n.$$

Proof: Let A be a set of n elements. Then, we know from the bijection rule & the product rule that

$$|\text{pow}(A)| = 2^n.$$

Now, let us see another way to find $|\text{pow}(A)|$. We know

$$\text{pow}(A) = \{B : B \subseteq A\}.$$

for $i \in \{0, \cdots, n\}$, let S_i be the set of all subsets of A containing exactly i elements. Then, from the subset rule, $|S_i| = \binom{n}{i}$. Note that

$$\text{pow}(A) = S_0 \cup S_1 \cup \cdots \cup S_n$$

As $S_0, \cdots, S_n$ are disjoint sets, it follows from the sum rule that

$$|\text{pow}(A)| = \sum_{i=0}^{n} \binom{n}{i}.$$

As $|\text{pow}(A)| = 2^n$, we get that

$$2^n = \sum_{i=0}^{n} \binom{n}{i}.$$

■

Remark: The algebraic proof for this claim is straightforward - substitute $a=b=1$ in the Binomial theorem.

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7. Basic Graph Theory

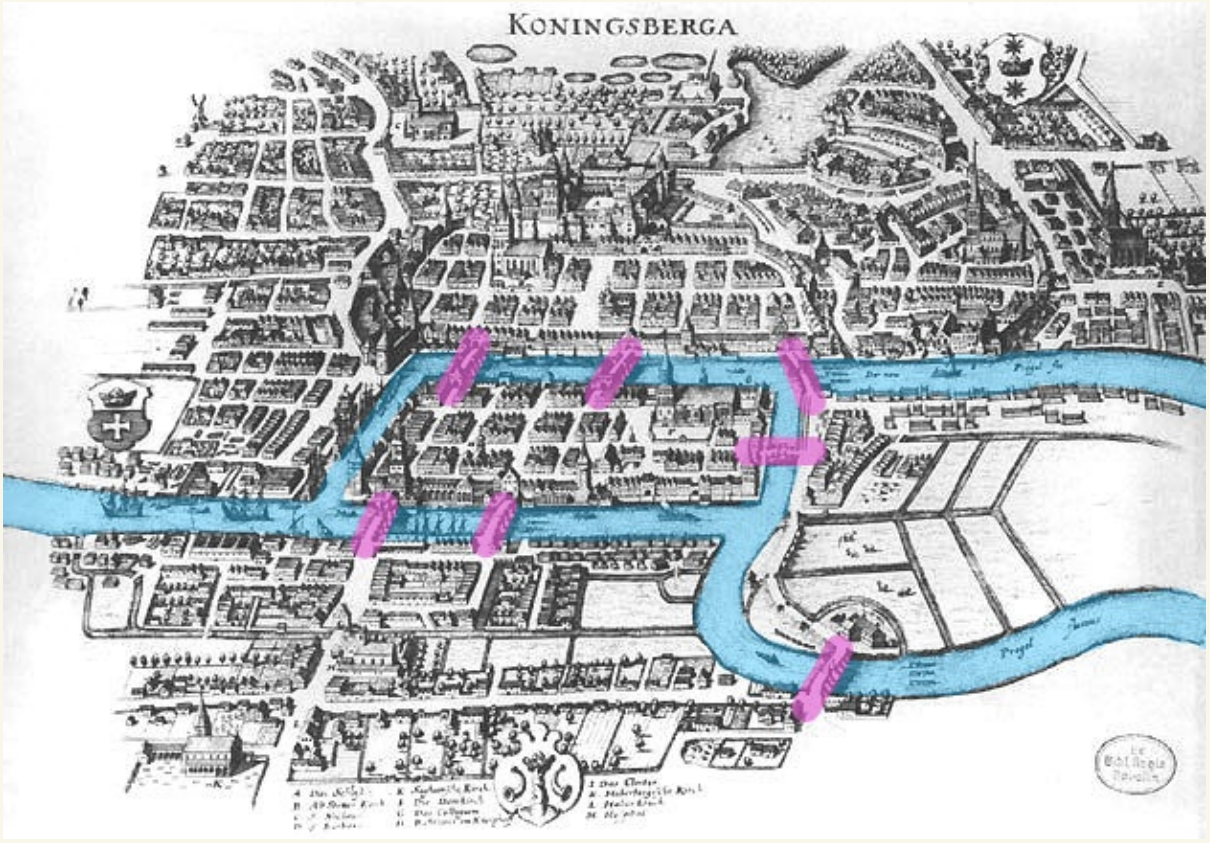
Graphs are almost everywhere - social network, metro map, UPI, etc. Graphs are heavily used in computer science - algorithms, machine learning, computer networks, theory of

computation, databases, etc.

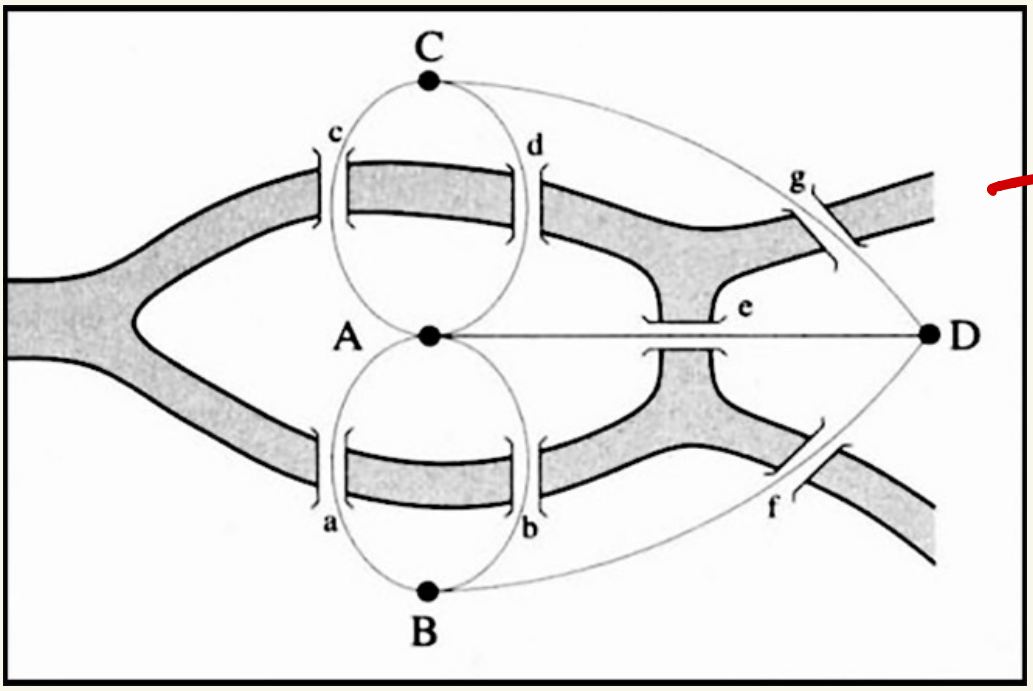
Let's start with a real-world problem that gave birth to graph theory.

The Königsberg Bridge Problem:

Königsberg is a city in Prussia located on Pregel river. The city had two islands and areas on both banks. The regions were linked by 7 bridges as shown below.



The inhabitants of Königsberg wondered whether they can leave their homes, cross every bridge exactly once & return back. The problem can be represented as follows



V = {A, B, C, D}

E = {a, b, c, d, e, f, g}

a = (A, B) = (B, A)

This figure helps us understand that the traversal desired by the inhabitants of Königsberg is not possible. This is so because each time one enters & leaves a land mass, they use two bridges connecting that landmass. Thus, the existence of the desired traversal requires that

every land mass must be connected to an even number of bridges, which is not the case with Königsberg bridges problem.

This problem led to the discovery of graph theory. The famous mathematician Leonhard Euler asked whether the 7 bridges of Königsberg could be crossed exactly once. He gave a negative answer in 1736, by considering the graphical

interpretation of the problem, as shown in the second figure above.

Now let's start with the formal definition of a graph.

Definition [Undirected graph]:

An undirected graph (or simply a graph) G is a pair $G=(V,E)$, where V & E are called the sets of vertices & edges of the graph G. Every edge in E has either one or two, but not more, vertices associated with it, called

its endpoints. An edge is said to connect its endpoints.

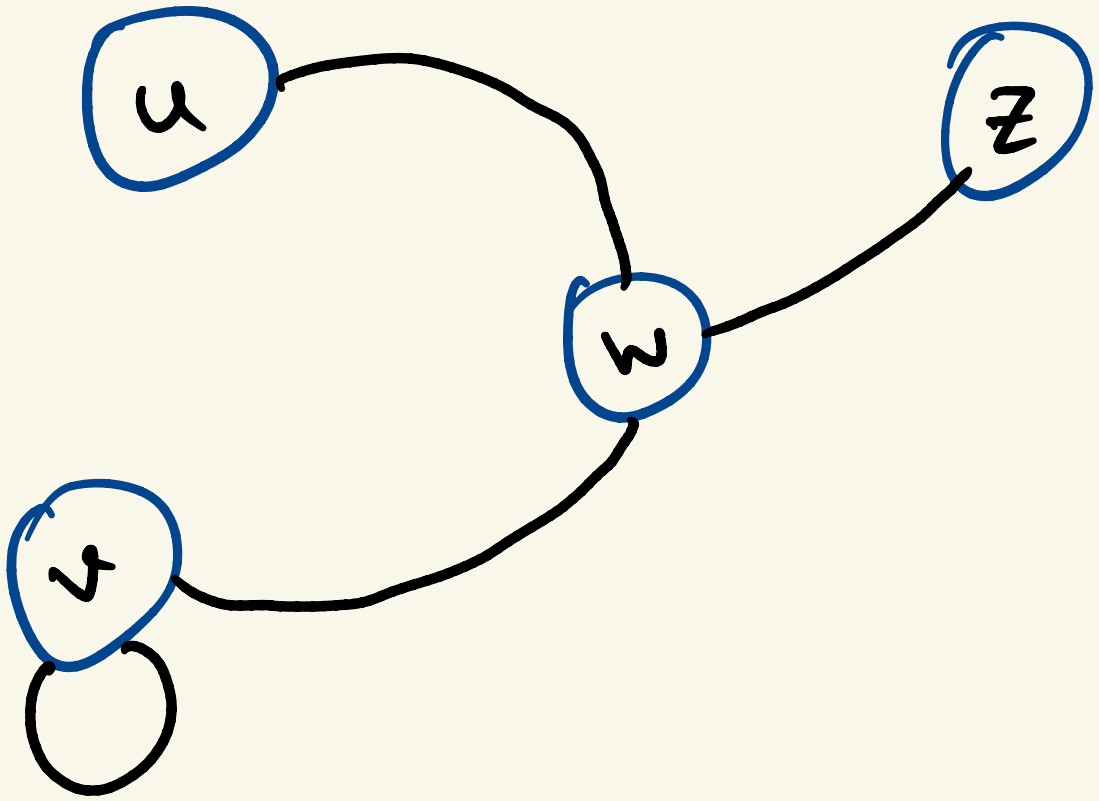
Eg.: i) In the second figure above, $V = \{A, B, C, D\}$ & $E = \{a, b, c, d, e, f, g\}$. The edge a has A & B as its endpoints, so we denote $a = (A, B)$. Similarly, for other vertices.

2) Let $V = \{u, v, w, z\}$ &

$$E = \{(u,w), (v,v), (u,w), (w,z)\}.$$

(u,w) (v,v) (w,u) (z,w)

The corresponding graph looks as follows:



24th November Lecture-25

Recap

a) Undirected graphs

7.1 Important definitions [For undirected graphs]

Let $G=(V,E)$ be an undirected graph.

1) Finite & infinite graphs: G is said to be finite if and only if both

V & E are finite. Otherwise, it is called an infinite graph. In this section, we will only deal with finite graphs.

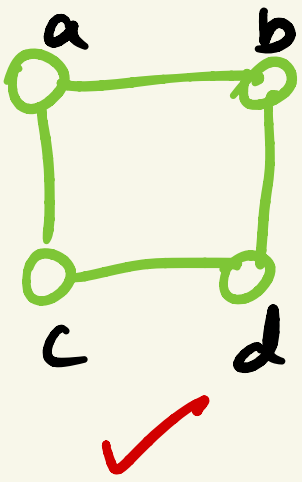
2) **Adjacent vertices**: Two vertices $u, v \in V$ are said to be <u>adjacent</u> if there exists an edge $e \in E$ such that u & v are endpoints of e. In this case, $e = (u, v)$.

3) **Incident edges**: We say that an edge $\overrightarrow{e}$ is incident to a vertex v if and only if v is an endpoint of e.

4) **Loop**: An edge $e \in E$ is called a loop if and only if e has the

same end points, i.e., $\exists\, v \in V$ such that $e = (v, v)$.

5) **Simple graphs**: G is said to be simple iff G has no loops & no multiple edges between any two vertices.



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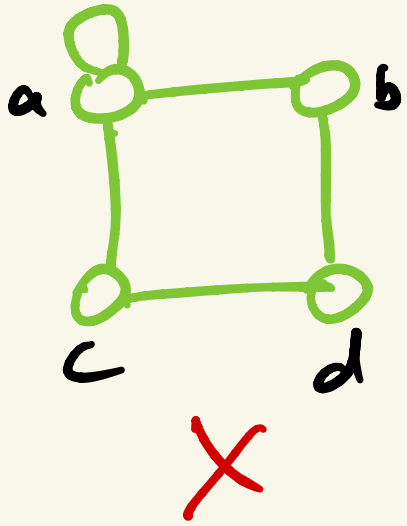

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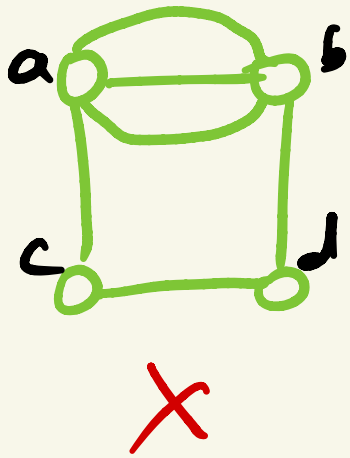

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6. **Null graph**: G is called the null graph if and only if $V = \phi$ & $E = \phi$.

7. **Neighborhood**: Let $v \in V$. Then, the neighborhood of v, denoted $N(v)$, is

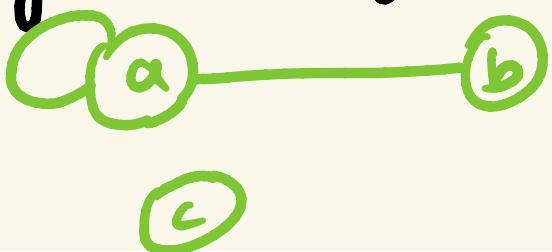
the set of all vertices adjacent to v in G. In other words,

$$\mathcal{N}(v) := \{u \in V \mid (u,v) \in E\}.$$

Let $A \subseteq V$. Then, the neighborhood of A, denoted $\mathcal{N}(A)$ is defined as

$$\mathcal{N}(A) := \bigcup_{v \in A} \mathcal{N}(v).$$

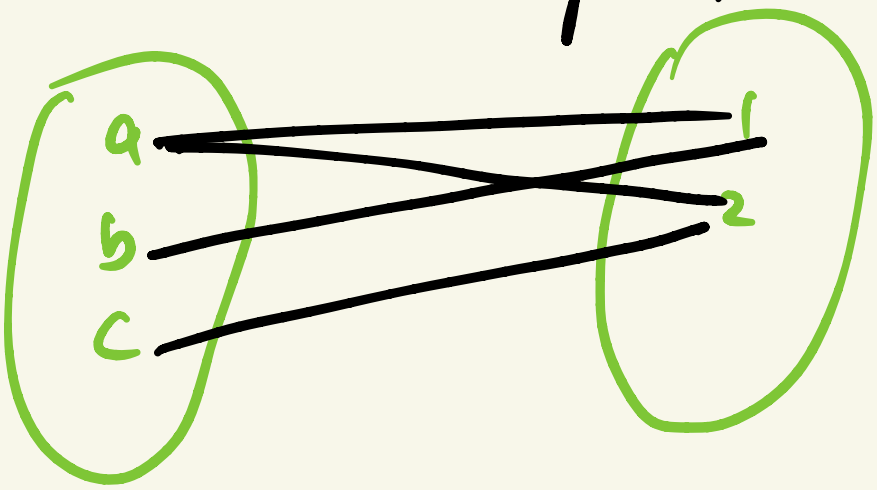
8. **Degree of a vertex**: Let $v \in V$. Then, the degree of v, denoted $\deg(v)$, is defined as the number of edges incident to v in G, except that a loop at v contributes twice to its degree. E.g.,



$\deg(a) = 3$
$\deg(b) = 1$
$\deg(c) = 0$

9. **Isolated vertex**: A vertex v in G is said to be isolated if and only if $\deg(v) = 0$.

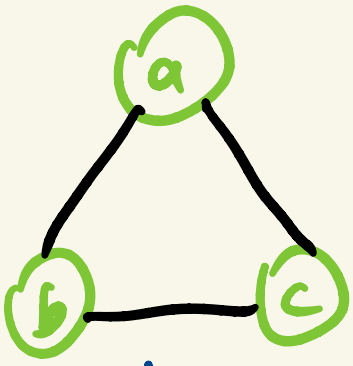
10. **Bipartite graph**: G is said to be bipartite if and only if there exist $V_1, V_2 \subseteq V$ such that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \phi$, and for every edge $e \in E$, there exist $v_1 \in V_1$ and $v_2 \in V_2$ such that v_1, v_2 are end points of e. For example,



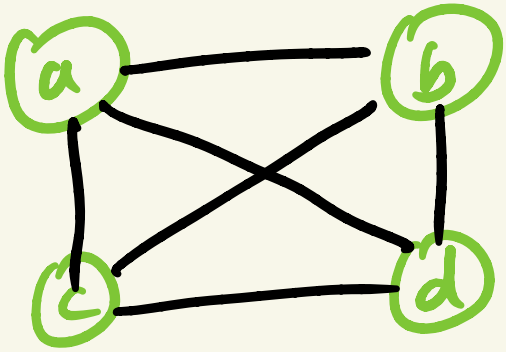
11. **Line graph**: G is said to be a line graph if and only if $|E| = |V| - 1$

for example,

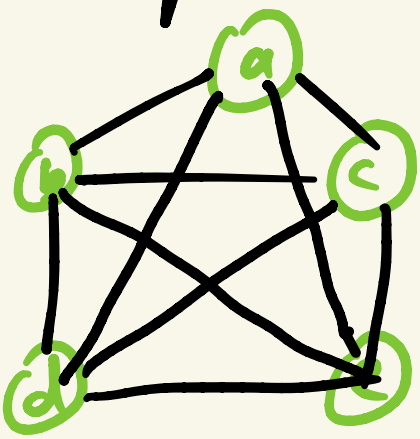
12. **Complete graph**: G is said to be the complete graph, if and only if for every $u, v \in V, u \neq v$, $(u, v) \in E$. The complete graph on n vertices is denoted by K_n. for example,



K_3



K_4



K_5

13. **Subgraph**: A graph $H = (U, F)$ is called a subgraph of $G = (V, E)$ if and only if $U \subseteq V$ and $F \subseteq E$.

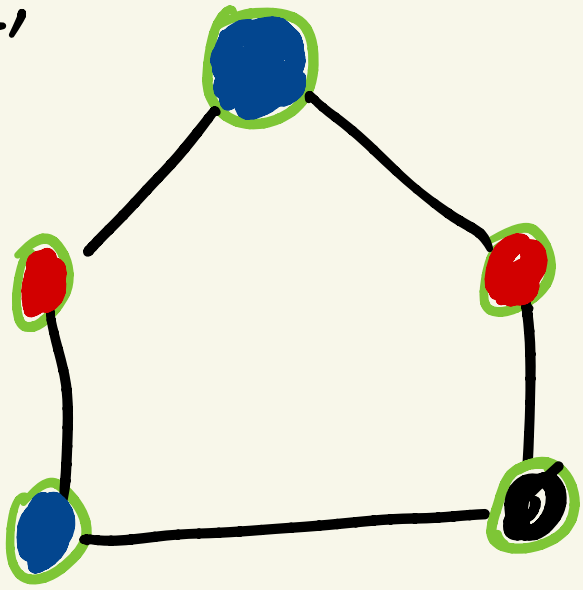
Further, H is said to be a proper subgraph of G if and only if $u \neq v$.

14. Complement of a graph: The complement of $G=(V,E)$, denoted $\bar{G}$, is the pair $(V,\bar{E})$, where for every $u,v \in V$, $\overrightarrow{uv} \in \bar{E}$ if and only if $(u,v) \notin E$. For example,

G $\qquad$ $\bar{G}$

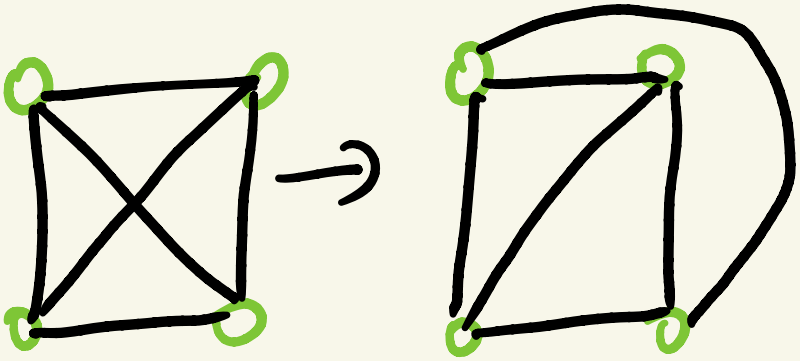
15. Chromatic number of a graph: The chromatic number of G, denoted χ_G, is the minimum number of colors needed to color the vertices

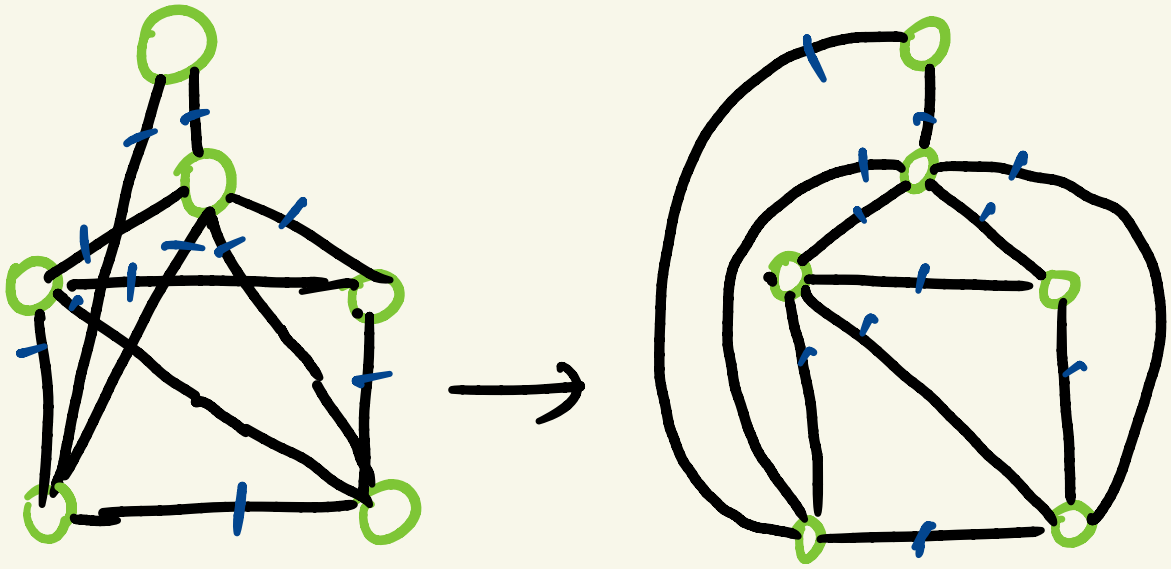
of G such that every adjacent vertices have different colors. for example,



$\chi_G = 3$

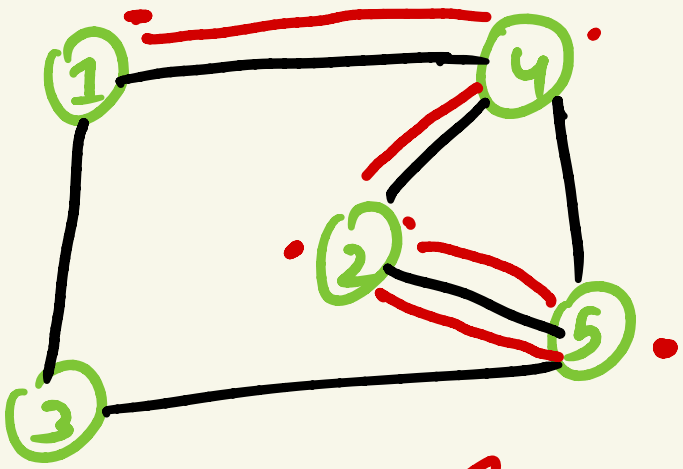
16. **Planar graph**: G is said to be planar if and only if it can be drawn in the plane without crossing the edges. For example,





17. **Walk in a simple graph**: Let $G=(V,E)$ be a simple undirected graph. A <u>walk</u> in G is an alternate sequence of vertices & edges, that begins with a vertex, ends with a vertex, such that for every edge $e=(u,v)$ in the walk, one of the endpoints in e is before it & the other endpoint in e is after it in the walk.

The length of a walk is the total occurrences of edges in it.



1-(1,4)-4-(4,2)-2-(2,5)-5-(2,5)-2

is a walk of length 4.

The vertices & edges can repeat in a walk. There are different types of walks in G.

a) Path: A walk in G is a path if and only if all vertices appearing in it are distinct.

For example, 1-(1,4)-4-(4,2)-2 is a path in the figure above, but 4-(4,2)-2-(2,5)-5-(5,4)-4

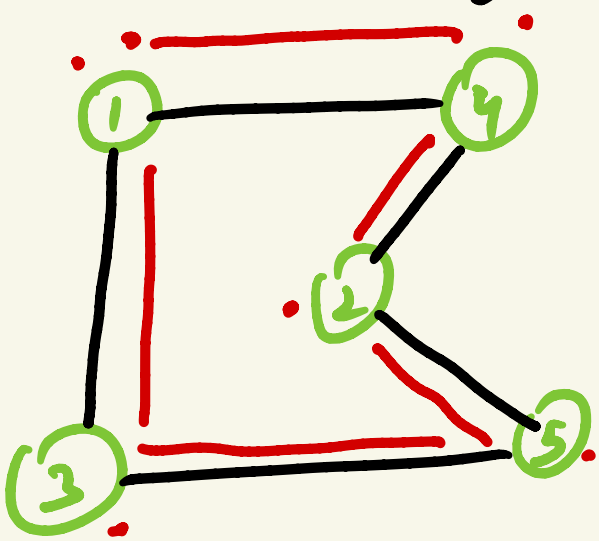
is not a path.

b) **Closed walk**: A walk in G that begins & ends at the same vertex is called a closed walk. For example, $2-(2,4)-4-(4,2)-2$ is a closed walk.

c) **Cycles**: A closed walk in G is said to be a cycle iff it has length at least three and all the vertices appearing in it except the beginning & end vertices are distinct. For example, $4-(4,2)-2-(2,5)-5-(5,4)-4$ is a cycle but $2-(2,4)-4-(4,2)-2$ is not.

d) Euler tour: Named after Leonhard Euler.

An Euler tour (also called Euler circuit) in G is a closed walk that includes every edge of G exactly once. For example,



1 - (1,4) - 4 - (4,2) - 2 - (2,5) - 5 - (5,3) - 3 - (3,1) - 1 is an Euler tour.

e) Hamiltonian cycles & paths:

A Hamiltonian cycle in G is a cycle such that every vertex except the starting & the end vertex appear exactly once & starting & end vertex

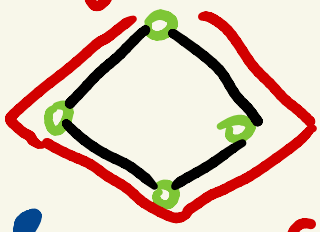
twice. Similarly, Hamiltonian path is a path with the same property.

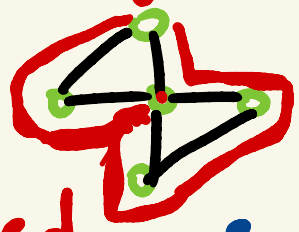
The example given in the previous point is also a Hamiltonian walk & it appears that Euler & Hamiltonian walks are the same, which is not true. Consider the following

Hamiltonian cycle

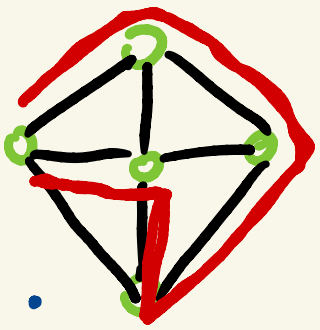
Not Hamiltonian cycle

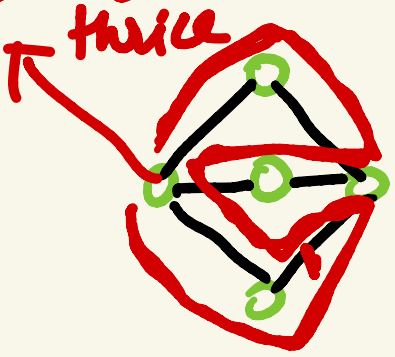
Euler tour





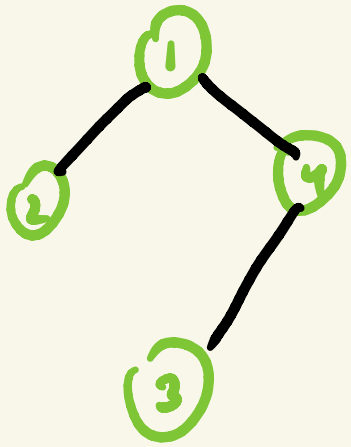
covered twice





Not Euler tour

18. **Acyclic graph**: G is said to be acyclic if and only if G does not have cycles. For example,



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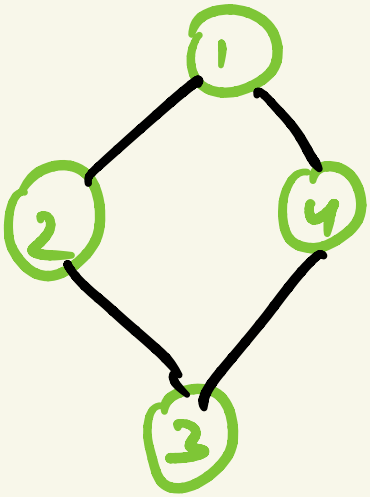
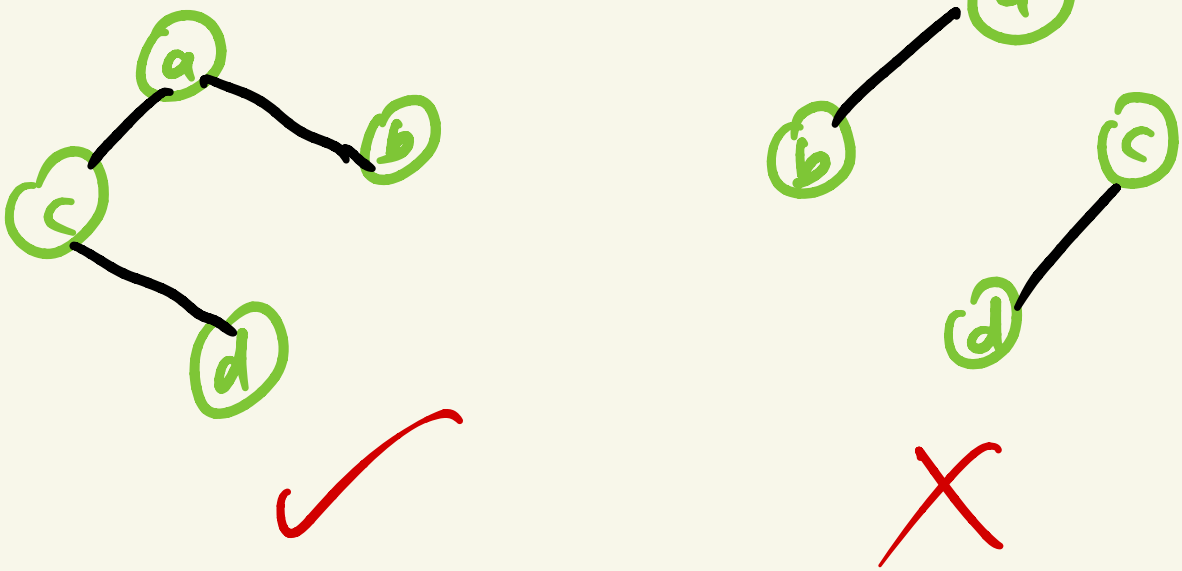


✗

19. **Connected graphs**: Two vertices u, v in G are said to be connected if there exists a path in G that begins at one of the two vertices & ends at the other.

We say that G is connected if and only if every pair of

vertices in G is connected.
for example,



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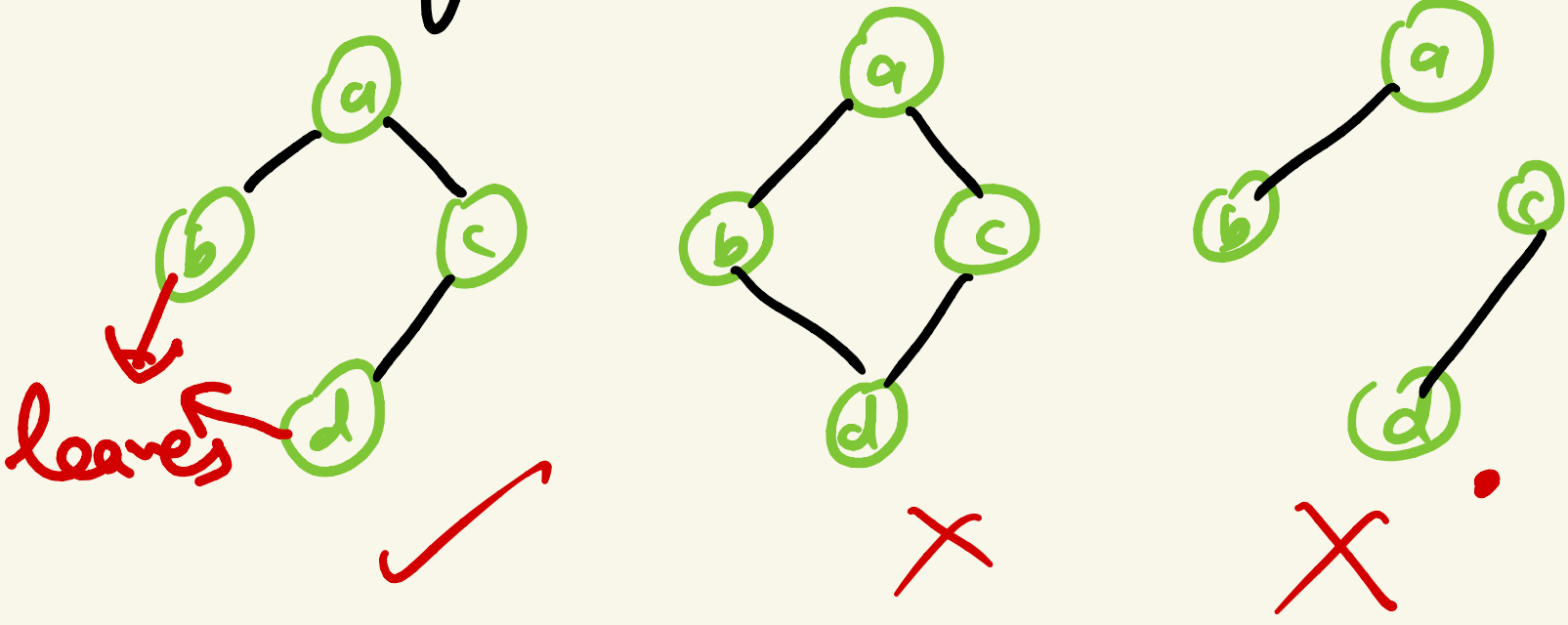
19. Trees: An undirected graph G is called a tree if and only if G is connected and G has no cycles. for example



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- A vertex of degree one in a tree is called a leaf.
- An acyclic graph is also called a forest. Every tree is a forest but not the other way around. For example, the third figure above is a forest but not a tree.

7.2 Important results:

A) Degree of vertices

Lemma 7.2.1 [The Handshaking Lemma]: Let $G=(V,E)$ be an undirected finite graph & $|E|=m$. Then,

$$\sum_{v \in V} \deg(v) = 2m.$$

Proof: We know that for every $v \in V$, $\deg(v)$ is equal to sum of edges incident on v. (counting loops twice) Let e be an edge incident on v. Then, there are two cases: either e is a loop on v, i.e., $e=(v,v)$ or $e=(v,u)$ for some $u \in V-\{v\}$. In the former case, the loop e contributes 2 to $\deg(v)$ and in the latter case, the edge e contributes 1 to $\deg(v)$ and 1 to $\deg(u)$. This implies that each edge in G contributes 2 to $\sum_{v \in V} \deg(v)$. Hence,

$$\sum_{v \in V} \deg(v) = 2m.$$

Lemma 7.2.2: Let $G=(V,E)$ be an undirected graph. Then, G has an even number of vertices of odd degree.

Proof: Let V_1 be the set of all vertices in G having odd degree and let $V_2 := V - V_1$. Then, we know $V_1 \cap V_2 = \phi$ and $V = V_1 \cup V_2$. Then,

$$\underbrace{\sum_{v \in V} \deg(v)}_{\cdot\, 2|E|} = \underbrace{\sum_{v_1 \in V_1} \deg(v_1)}_{\text{even}} + \underbrace{\sum_{v_2 \in V_2} \deg(v_2)}_{\cdot\, \text{even}}.$$

We know from Lemma 7.2.1 that

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Also, as every vertex in V_2 has

even degree, we get that

$$\sum_{v_2 \in V_2} \deg(v_2) = 2k$$

for some $k \in \mathbb{N}$. Then, we get that there exists a $b \in \mathbb{N}$ such that

$$\sum_{v_1 \in V_1} \deg(v_1) = 2b.$$

We know that every vertex in V_1 has odd degree Then, the above equation that $|V_1|$ is even. ■

↳ implies

B. Matchings

Definition [Matching & perfect matching]: Let $G = (V, E)$ be an undirected graph. A matching M of G is a

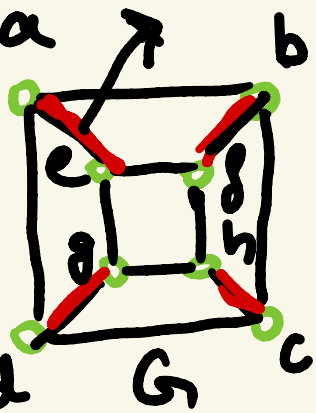
subset of E such that M does not contain loops and no two edges in M have a common endpoint.

The vertices incident to edges in M are said to be saturated by M & other vertices are said to be M-unsaturated.

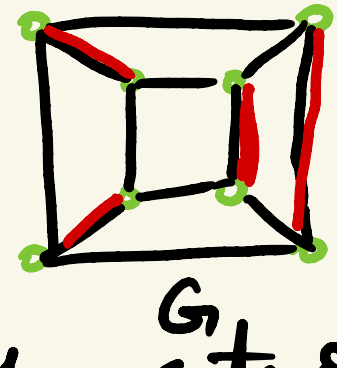
- M is called a perfect matching of G if and only if every vertex in G is saturated by M.

E.g.:

1)



2)



In these examples, the sets of red edges are two perfect matchings of G.

One may wonder if every graph having "enough" edges also have a perfect matching.

Claim 7.2.3: Let $n \in \mathbb{N} - \{0\}$ and K_{2n+1} be the complete graph on $2n+1$ vertices. Then, K_{2n+1} does not contain a perfect matching.

Proof: We know that K_{2n+1} contains odd number of vertices. Let M be an arbitrary matching on K_{2n+1}. As no pair of edges in M has a common endpoint, it follows immediately that M saturates $2|M|$ many vertices. Thus, for every matching M on K_{2n+1},

there exists an M-unsaturated vertex in K_{2n+1}. Hence, no matching M on K_{2n+1} is a perfect matching. ∎

We can ask a similar question for a complete graph having even number of vertices.

Claim 7.2.4: Let $n \in \mathbb{N} - \{0\}$ & K_{2n} be the complete graph on $2n$ vertices. Then, K_{2n} has $\frac{(2n)!}{2^n \cdot n!}$ many perfect matchings.

Proof: Let p_{2n} be the number of perfect matchings in K_{2n} and $V = \{v_1, \cdots, v_{2n}\}$ be the set of

vertices of K_{2n}. Note that p_{2n} is the number of ways to make n pairs of $v_1, \cdots, v_{2n}$, where the collection of these n pairs corresp- ond to a perfect matching of K_{2n}. Let $\sigma : \{1, \cdots, 2n\} \longrightarrow \{1, \cdots, 2n\}$ be a permutation. Consider the ordered set V_σ defined as follows

$$V_\sigma := (v_{\sigma(1)}, v_{\sigma(2)}, \cdots, v_{\sigma(2n-1)}, v_{\sigma(2n)}).$$

Let

$$M_\sigma := \{(v_{\sigma(1)}, v_{\sigma(2)}), (v_{\sigma(3)}, v_{\sigma(4)}), \cdots, (v_{\sigma(2n-1)}, v_{\sigma(2n)})\}.$$

As K_{2n} is a complete graph, $(v_{\sigma(i)}, v_{\sigma(i+1)})$ is an edge of

K_{2n} for every $i \in \{1,3,5,\cdots,2n-1\}$.
This implies M_σ is a perfect matching in G_1. Let

$$P := \{V_\sigma := (v_{\sigma(1)},\cdots,v_{\sigma(2n)}) \mid \sigma \text{ is a permutation on } \{1,2,\cdots,2n\}\}.$$

Let $\mathcal{M}$ be the set of all perfect matchings on K_{2n}. Then, it is not difficult to see that $\mathcal{M}$ looks as follows ← HW

$$\mathcal{M} = \{M_\sigma : \sigma \text{ is a permutation on } \{1,\cdots,2n\}\},$$

where $M_\sigma = \{(v_{\sigma(1)}, v_{\sigma(2)}),\cdots,(v_{\sigma(2n-1)}, v_{\sigma(2n)})\}$

Let $F: P \to M$ be the function defined as follows: for every permutation σ on $\{1, \cdots, 2n\}$, $F(V_\sigma) := M_\sigma$.

Let us take an arbitrary $M_\sigma \in M$. We want to count the number of pre-images of M_σ under F.

We know that $F(V_\sigma) = M_\sigma$.

Recall $V_\sigma = (V_{\sigma(1)}, V_{\sigma(2)}, \cdots, V_{\sigma(2n-1)}, V_{\sigma(2n)})$ is an ordered set and

$$M_\sigma = \{(V_{\sigma(1)}, V_{\sigma(2)}), \cdots, (V_{\sigma(2n-1)}, V_{\sigma(2n)})\}$$

is an unordered set.

We can also view V_σ as

$$V_\sigma = (\{V_{\sigma(1)}, V_{\sigma(2)}\}, \cdots, \{V_{\sigma(2n-1)}, V_{\sigma(2n)}\})$$

and regard it as an ordered set of n pairs. Note that if we

either permute these n pairs or swap the positions of two elements in any pair, and then apply f to the resulting order set, we get M_σ. It follows from this discussion that f is a $(2^n \cdot n!)$-to-1 function. As $|P| = 2n!$, it follows from the division rule that

$$|M| = \frac{(2n!)}{2^n \cdot n!}.$$

■

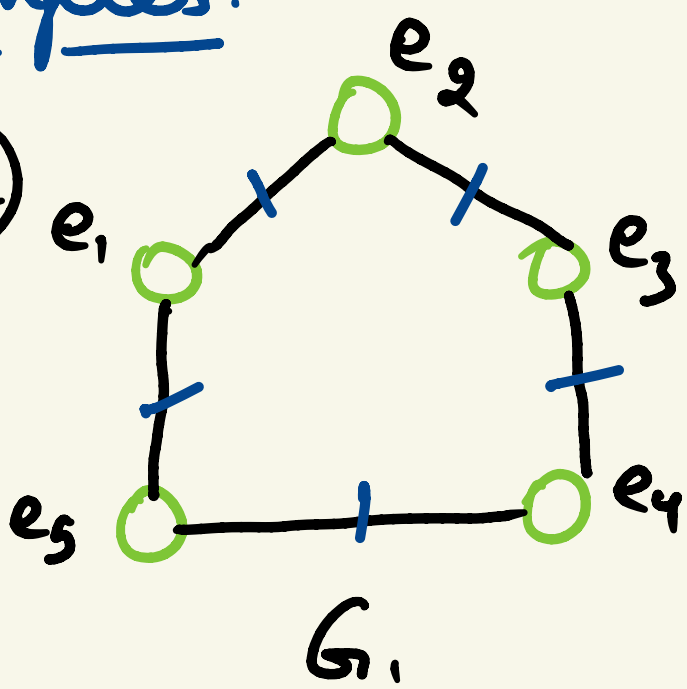
C. Graph Isomorphism

Definition [Isomorphic graphs]: Two finite undirected graphs

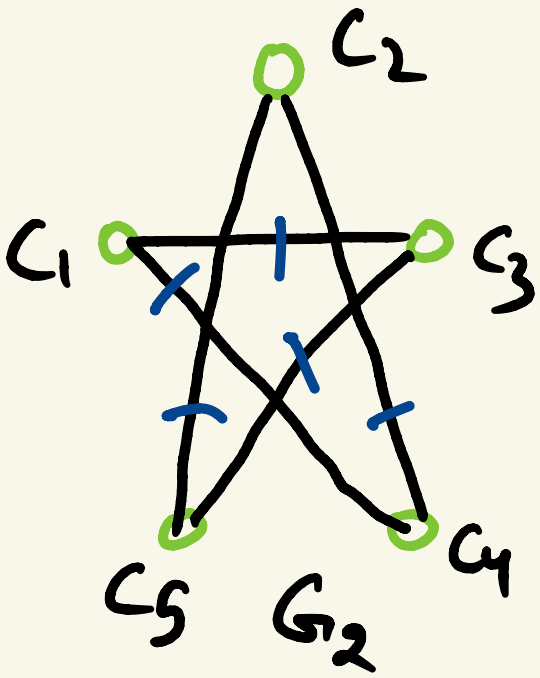
$G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be isomorphic if and only if there exists a bijective function $\pi : V_1 \to V_2$ such that for every $u, v \in V_1$, $(u, v) \in E_1$ if and only if $(\pi(u), \pi(v)) \in E_2$.

Examples:

1)



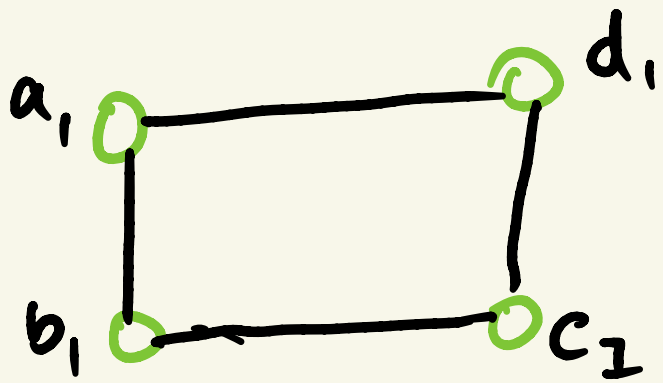
G_1



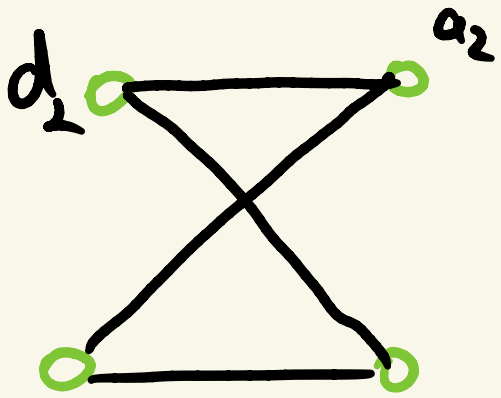
G_2

G_1 & G_2 are isomorphic : $\pi : V_1 \to V_2$ is as follows: $\pi(e_1) = c_1$, $\pi(e_2) = c_3$, $\pi(e_3) = c_5$, $\pi(e_4) = c_2$, $\pi(e_5) = c_4$.

2)



G_1



G_2

G_1 & G_2 are isomorphic. The bijective function $\sigma: V_1 \to V_2$ is as follows: $\sigma(a_1) = a_2$, $\sigma(b_1) = b_2$, $\sigma(c_1) = c_2$, $\sigma(d_1) = d_2$.

We say that a graph property is preserved under graph isomorphism if and only if whenever G satisfies that property, all graphs isomorphic to G also

satisfy that property. We now list some properties of graphs which are preserved under graph isomorphism. Let $G_1=(V_1,E_1)$ and $G_2=(V_2,E_2)$ be two isomorphic graphs and $\pi: V_1 \to V_2$ be a graph isomorphism from G_1 to G_2. Then,

HW 1) G_1 & G_2 have the same number of vertices & the same number of edges.

HW 2) for every $v \in V_1$, $u \in N(v)$ if and only if $\pi(u) \in N(\pi(v))$ where $N(v)$ is the neighborhood of v in G_1.

HW 3) for every $v \in V_1$, $\underbrace{\deg(v)}_{\text{in } G_1} = \underbrace{\deg(\pi(v))}_{\text{in } G_2}$

Graph isomorphism plays a very important role in computer science. In particular, testing whether two graphs are isomorphic or not has been studied for more than six decades & yet we do not have a polynomial-time algorithm for it. The sanity check in algorithms for graph isomorphism is to check if all the graph properties mentioned above are satisfied or not.

If they are not satisfied, we get that the two given graphs are not isomorphic. However, if they are satisfied, it does not mean that the two given graphs are isomorphic.

26th November Lecture 26

Recap:

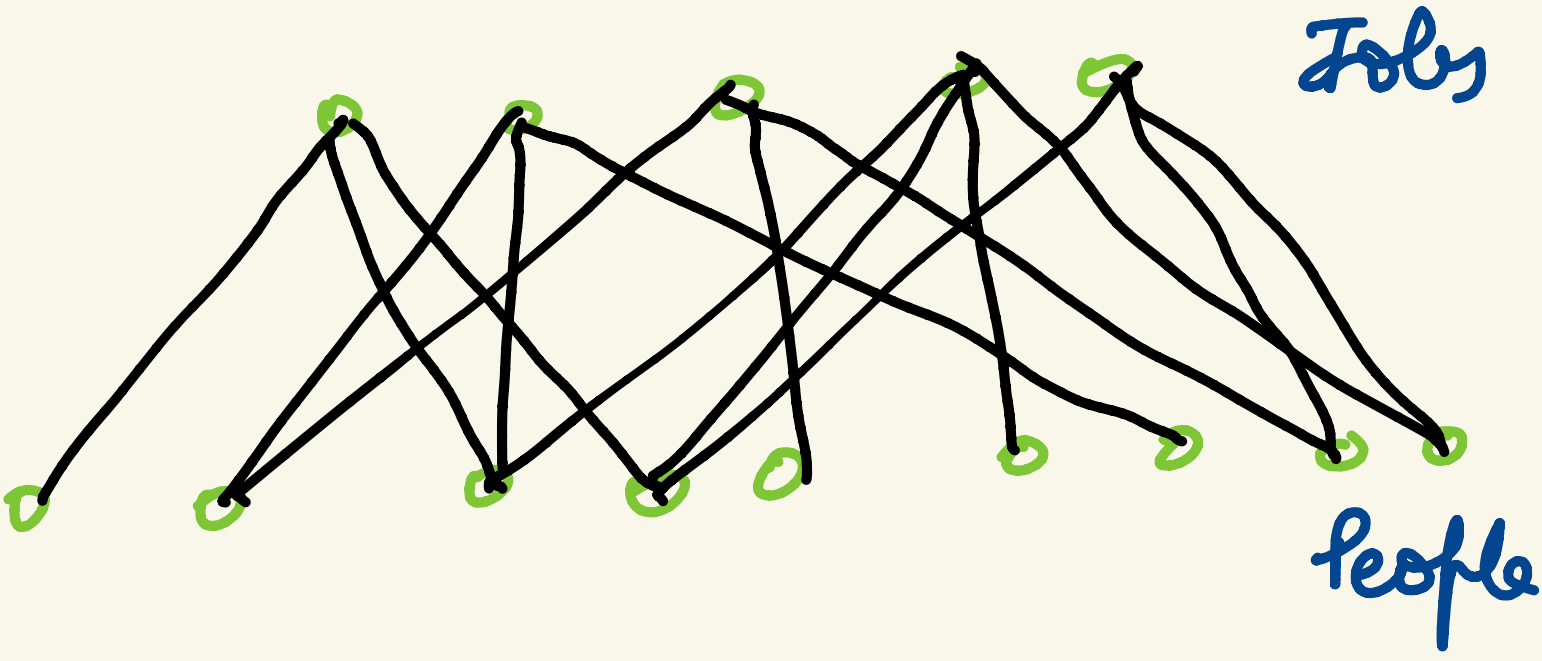
a) undirected graphs
b) Various definitions
c) Handshaking lemma
d) Matchings and perfect matchings

c) Number of perfect matchings in complete graphs

D) Matchings in bipartite graphs

- **Job assignment & bipartite graphs:** Suppose there are m jobs and n people, $n \geq m$; but not everyone is qualified for all jobs. Can all jobs be assigned? We model this as a bipartite graph as follows: Let V_1 be

the set of all m jobs & V_2 be the set of all n people. for $v_1 \in V_1$ and $v_2 \in V_2$, $(v_1, v_2) \in E$ if and only if v_2 is qualified for job v_1. Let $V = V_1 \cup V_2$ & $G = (V, E)$. Then, G models the problem of job assignment. for example,



we want that each job must be assigned to exactly one person and every person can hold at most one job. In other words, we are looking for a matching in the graph G. We need the following definition.

Definition [Complete matching]:

Let $G=(V,E)$ be a bipartite graph and $V=V_1 \cup V_2$, where $V_1 \cap V_2 = \phi$. Assume $|V_1| \leq |V_2|$. A set $M \subseteq E$ is called

a complete matching if and only if M is a matching in G & $|M| = |V_1|$.

- If $|V_1| = |V_2|$ then a complete matching is also a perfect matching.
- In the above example, we can replace jobs & people with men & women and ask the following question related to marriage: map a man to a woman if both of them are

eligible to marry each other. Assuming we have an equal number of men & women, we can ask if we can find a mapping such that everyone gets married.

The following result answers such questions. This is a celebrated result proved by Philip Hall in 1935.

Theorem 7.2.5 [Hall's marriage theorem]: Let $G = (V, E)$ be a

bipartite graph such that $V = V_1 \cup V_2$ where $V_1 \cap V_2 = \phi$. Assume $|V_1| \leq |V_2|$. Then, G has a complete matching if and only if for every $A \subseteq V_1$, $|N(A)| \geq |A|$, where $N(A)$ is the neighbourhood of vertices in A.

$$N(A) = \bigcup_{v \in A} N(v)$$

Proof: Suppose G has a complete matching M. This means that for every vertex $v_1 \in V_1$, there exists a vertex $v_2 \in V_2$

such that $(v_1, v_2) \in M$. Let $A \subseteq V_1$ be an arbitrary subset. As M is a complete matching, for every $v_1 \in A$, there exists a $v_2 \in V_2$ such that $(v_1, v_2) \in M$ and for every $v_1' \in V_1 - \{v_1\}$, $(v_1', v_2) \notin M$. Then, clearly $|N(A)| \geq |A|$.

Now, assume that for every $A \subseteq V_1$, $|N(A)| \geq |A|$. We want to show that there exists a complete matching in G. We prove this result by strong

induction on $|V_1|$. Let $|V_1| = m$.

a) **Base case:** Suppose $|V_1| = 1$, i.e., $V_1 = \{v_1\}$. We know that $|N(V_1)| \geq |N_1|$. Then, there exists a $v_2 \in V_2$ such that $(v_1, v_2) \in E$. Let $M = \{(v_1, v_2)\}$. Then, M is a complete matching in G. Hence, the base case is true.

b) **Induction hypothesis:**

Let $G' = (V', E')$ be a bipartite graph such that $V' = V_1' \cup V_2'$, where $V_1' \cap V_2' = \phi$

and $|V_1'| < m$. Assume that for every $A' \subseteq V_1'$, $|N(A')| \geq |A'|$.
assume that
Then, every such G' has a complete matching.

c) Induction step: We want to prove that G has a complete matching. We prove this by cases.

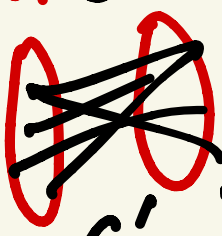
1) Case 1: Suppose for every $k \in \{1, \dots, |V_1|\}$, every subset $A \subseteq V_1$, satisfying $|A| = k$, we have $|N(A)| > |A|$. Let $v_1 \in V_1$ be arbitrary & $V_1' := V_1 - \{v_1\}$. Let $v_2 \in N(v_1)$.

We know $|N(V_1')| \geq |V_1'| \Rightarrow \geq |V_1'|+1$

As $|N(V_1')| \geq |V_1| = |V_1'| + 1$, we get that for every $A' \subseteq V_1'$, $|N(A' - \{v_2\})| \geq |A'|$.

AN

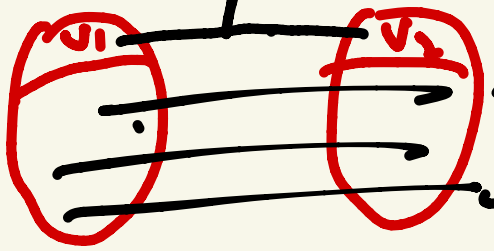
Consider the graph G' obtained by removing v_1 from V_1 and v_2 from V_2 and all edges in E which are either incident to v_1 or v_2. Note that G' is a bipartite graph where V_1' and $V_2 - \{v_2\}$ for a partition.



As noted above, for every $A' \subseteq V_1'$, $|N_{G'}(A')| \geq |A'|$, where $N_{G'}$ is neighborhood in G'.

Then, from the induction hypothesis, there exists a

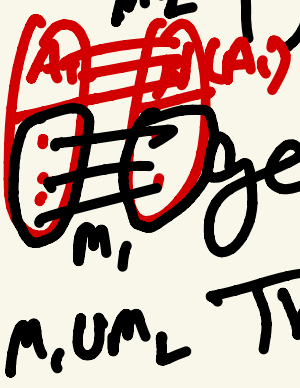
complete matching M' in G'. Let $M := M' \cup \{(v_1, v_2)\}$. Then, clearly, M is a complete matching in G.



2) Case 2: Suppose there exists $k \in \{1, \ldots, m\}$ and $A \subseteq V_1$ satisfying $|A| = k$ such that $|N(A)| = |A|$. Let $V_1' := V_1 - A$ and $V_2' := V_2 - N(A)$. Let $\tilde{G}$ be the graph whose vertices are $A \cup N(A)$ and whose edge set is the set of all edges between A & $N(A)$ in G. Then, $\tilde{G}$

is a bipartite graph. For $B \subseteq A$, we denote the neighborhood of B in $\tilde{G}$ by $N_{\tilde{G}}(B)$. Then, we claim that for every $B \subseteq A$, $|N_{\tilde{G}}(B)| \geq |B|$. Suppose not, i.e., there exists a $B \subseteq A$ such that $|N_{\tilde{G}}(B)| < |B|$. Then it follows that $|N(B)| < |B|$, where $N(B)$ is the neighbourhood of B in G. This is so because all edges incident to vertices in B are present in $\tilde{G}$ & hence $N(B) \subseteq N(A)$.

$|N(B)| < |B|$ can't happen as by assumption, for every $B' \subseteq V_1$, $|N(B')| \geq |B'|$. Hence, we get that $|N_{\tilde{G}}(B)| \geq |B|$.



Thus, we get from the induction hypothesis that there exists a complete matching $\tilde{M}$ in $\tilde{G}$.

Let G' be the graph obtained by removing A from V_1 & $N(A)$ from V_2 and removing all edges incident to vertices in $A \cup N(A)$. Clearly, G' is a bipartite

graph with vertex set $V_1' \cup V_2'$, where $V_1' := V - A$ & $V_2' := V_2 - N(A)$. Now, we claim that for every $A' \subseteq V_1'$, $|N_{G'}(A')| \geq |A'|$, where $N_{G'}(A')$ is the neighborhood of A' in G'. Suppose this is not true, i.e., there exists an $A' \subseteq V_1'$ such that $|N(A')| < |A'|$. Then, as $|N(A)| = |A|$ & $A' \cap A = \phi$, we get that $|N(A' \cup A)| < |A \cup A'|$, where $N(A' \cup A)$ is the neighborhood of $A' \cup A$ in G. This is a contradiction to

the assumption that we have made. Now it follows from the induction hypothesis that there exists a complete matching M' in G'.

Let $M := \widehat{M} \cup M'$. Then, it follows that M is a complete matching in G. ∎

7.3 Directed graphs

These are the graphs where every edge has a direction. More formally,

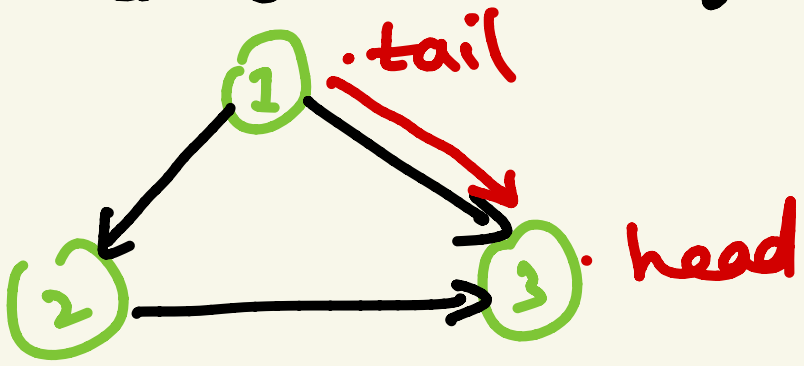
Definition [Directed graphs]: A directed graph (or digraph) is a pair

(V,E), where $V \neq \phi$ is the set of vertices & E is the set of directed edges of G. A directed edge, denoted $\langle u,v \rangle$, starts at the vertex u & ends at the vertex v. In this case, u is called the tail & v is called the head of edge.

- Example:

$V = \{1,2,3\}$

$E = \{\langle 1,2\rangle, \langle 2,3\rangle, \langle 1,3\rangle\}$

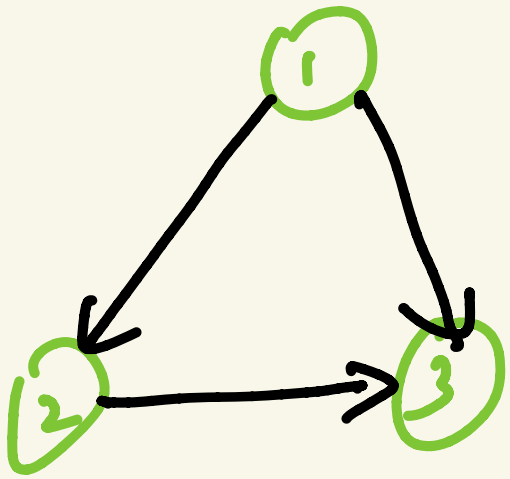


Definition [In-degree & out-degree of vertices]: Let $G=(V,E)$ be a digraph & $v \in V$ be arbitrary.

Then, $\text{in-degree}(v) := |\{e \in E : v \text{ is the head of } e\}|$

4 out-degree$(v) := |\{e \in E : v \text{ is the tail of } e\}|$

Example:



in-deg$(1) = 0$,
out-deg$(1) = 2$,
in-deg$(2) = 1$,
out-deg$(2) = 1$,
in-deg$(3) = 2$
out-deg$(3) = 0$.

Claim 7.3: Let $G = (V, E)$ be a finite diagraph. Then,

$$\sum_{v \in V} \text{in-deg}(v) = \sum_{v \in V} \text{out-deg}(v)$$

HW

Proof idea: Every edge in G corresponds 1 to $\sum_{v \in V} \text{indeg}(v)$

& 1 to $\sum_{v \in V}$ out-deg(v). ■

Remark: We can define almost all definitions given in Section 7.2 also for digraphs.

7.4 Trees

Recall that a tree is a connected acyclic undirected graphs. Trees are extensively used in mathematics & computer science, for example data structures, algorithms, storage, etc.

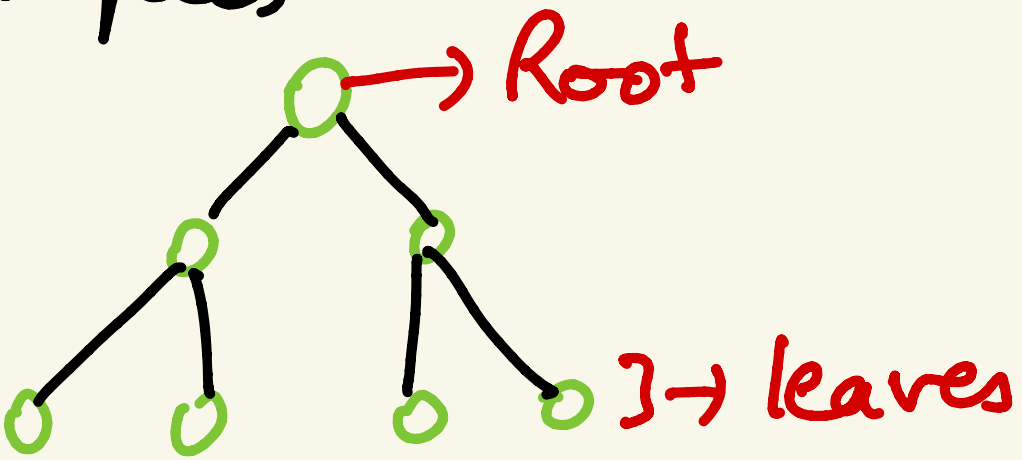
- Recall that a vertex of degree 1 in a tree is called a

leaf & other vertices are called as internal vertices.

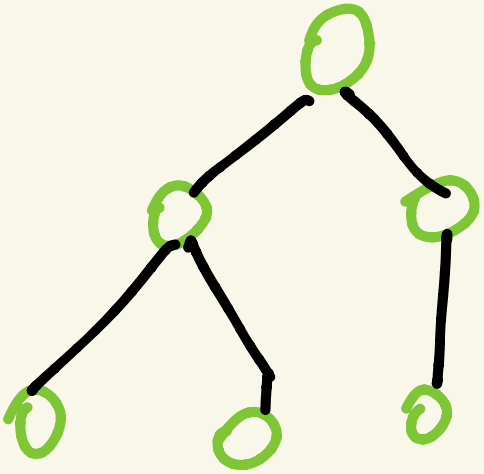
Definition [m-ary trees]: Let $m \in \mathbb{N} - \{0\}$ and $G = (V, E)$ be a tree. Then, G is called an m-ary tree if and only if every vertex in G has degree at-most $m+1$. If $m = 2$, G is called a binary tree.

- If every internal vertex has degree exactly m, G is called a complete m-ary tree. If $m = 2$ then G is called a complete binary tree.

Examples:

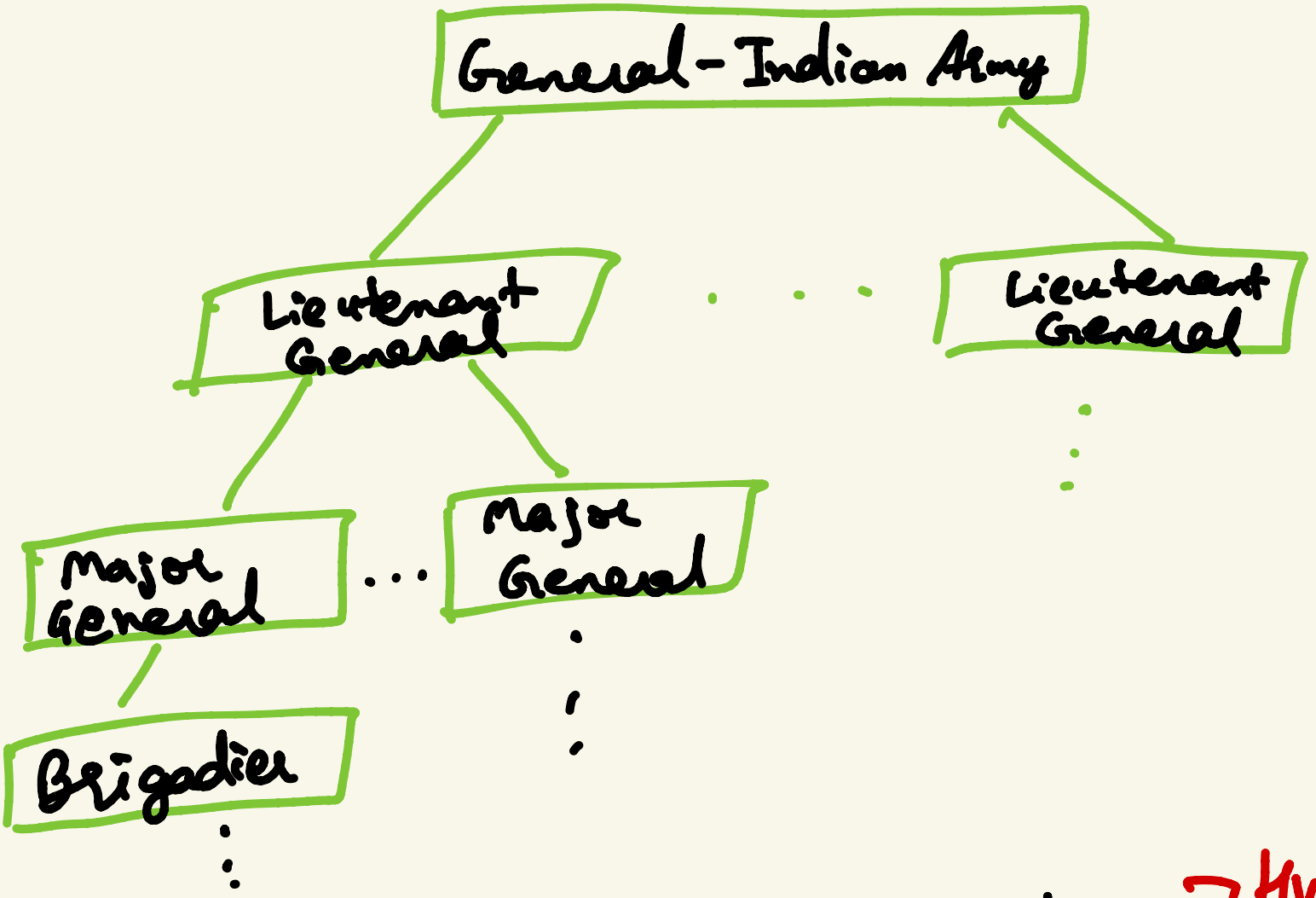


Complete binary tree



Binary tree (but not complete).

- Trees are used to represent hierarchies.



Claim 7.4.1: Let G be a tree with n vertices. Then, G has $n-1$ edges. HW

Proof [By induction on n]:

a) **Base case:** Let $n=1$. Then, G has 0 edges & the claim holds.

b) Induction hypothesis: Let $1 < k < n$ be an arbitrary natural number and G' be a tree having k vertices. We assume that G' has $(k-1)$ edges.

c) Induction step: Let v be an arbitrary leaf node & u be the only neighbour of v in G (also called the parent of v). We remove v & the edge connecting u & v from G. Assume G has $(k+1)$ vertices. After removing v & the edge connecting u & v from G, we are left with k vertices. By induction hypothesis, G' has k vertices

→ call this tree G',

& (k-1) edges. Thus, when we add v & the edge connecting u & v back to G', we get G. Hence, G has (k-1) edges. This proves the induction step. ∎

Homework: Read more about trees from the book Discrete Mathematics by Kenneth H. Rosen.